

• Marginalisation and Conditioning

$$f_{X,Y}(x,y) = f_{X|Y}(x|y) \cdot f_Y(y) \Rightarrow f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$f_Y(y) = \int f_{X,Y}(x,y) dx$$

Let $X \sim N(\mu, \Sigma)$.

Partition X as $[X_1, X_2]^T$

Similarly, $\mu = \begin{matrix} m \times 1 & (n-m) \times 1 \end{matrix} [\mu_1, \mu_2]^T$ and $\Sigma = \begin{matrix} m \times m \end{matrix} \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$

Marginalisation Property:

The m -dimensional marginal distribution of X_1 is $N(\mu_1, \Sigma_{11})$ and X_2 is $N(\mu_2, \Sigma_{22})$.

$$f_{X_1, \dots, X_m}(x_1, \dots, x_m) = \int f_X(x) dx_{m+1} dx_{m+2} \dots dx_n$$

Conditioning Property:

The m -dimensional conditional distribution of X_1 given $X_2 = x_2$ is

$$N\left(\mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2), \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}\right)$$

note the decrease in variance
which does not depend on x_2 .

• Bivariate Gaussians

Suppose $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$.
and X_1 and X_2 are independent.

$$\bar{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \Rightarrow \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \Rightarrow |\Sigma| = \sigma_1^2 \sigma_2^2$$

$$\begin{aligned} f_{\bar{X}}(\bar{x}) &= f_{X_1, X_2}(x_1, x_2) \\ &= f_{X_1}(x_1) \cdot f_{X_2}(x_2) \\ &= \frac{1}{\sqrt{2\pi} \sigma_1} \cdot e^{-\frac{1}{2} \left(\frac{x_1 - \mu_1}{\sigma_1} \right)^2} \times \frac{1}{\sqrt{2\pi} \sigma_2} \cdot e^{-\frac{1}{2} \left(\frac{x_2 - \mu_2}{\sigma_2} \right)^2} \\ &= \frac{1}{(2\pi) \sigma_1 \sigma_2} \times e^{-\frac{1}{2} \left\{ \left(\frac{x_1 - \mu_1}{\sigma_1} \right)^2 + \left(\frac{x_2 - \mu_2}{\sigma_2} \right)^2 \right\}} \\ &= \frac{1}{(2\pi) \sqrt{\det(\Sigma)}} \cdot e^{-\frac{1}{2} \{ (x - \mu)^T \Sigma^{-1} (x - \mu) \}} \end{aligned}$$

* A vector composed of independent Gaussians is a Gaussian vector.

The converse is not true.

A vector of dependent Gaussian components need not be a Gaussian vector.

* Bivariate Gaussian is closed under marginalisation.

* Bivariate Gaussian is closed under conditioning.

• STOCHASTIC PROCESSES

A stochastic process $\{X(t), t \in T\}$ is a collection of random variables defined such that for every $t \in T$, we have $X(t): \Omega \rightarrow S$
↗ parameter space (often resembles time)
↘ state space

• Examples of Stochastic Processes

① Sequence $\{X_i\}$ of i.i.d random variables

General random walk

= If $X_1, X_2, \dots$ is a sequence of i.i.d random variables,
then $S_n = \sum_{i=1}^n X_i$ is a random walk.

1D random walks can have positive, negative or no drift depending on the sign of $E[X]$.

② Wiener Process / Brownian Motion

= $\{X(t), t \geq 0\}$
such that for every $t > 0$, $X(t) \sim N(0, t)$

③ Gaussian Process

A continuous time stochastic process $\{X_t, t \in T\}$ is a gaussian process iff for any finite set of indices $t_1 \dots t_k$, $[X_{t_1}, \dots, X_{t_k}]$ is a multivariate Gaussian vector.

④ $\{X_n, n \geq 0\}$ is a martingale if
 $E[X_{n+1} | X_1, \dots, X_n] = X_n$